

PYTHON TURTLE: DRAW WITH CODE

Grade 3 Coding Course

Think Unlimited

What is Python Turtle? Python Turtle is a fun drawing tool that lets you create pictures by writing code! You give commands like move forward or turn right, and the turtle draws lines on the screen. It is like having a tiny robot that follows your instructions! Code your imagination — one turtle step at a time!

Setting Up Your Turtle Friend

Let's Get Started! First, we need to import the turtle library and create our turtle friend. Type this code in your Python IDE and press **Run**!

```
import turtle
t = turtle.Turtle()
```

Basic Turtle Commands

Use these core commands to move your turtle and draw across the screen. Try them out and see what happens!

Command	What It Does	Visual Guide / Meaning
<code>t.forward(100)</code>	Move forward 100 steps	→ Moves the turtle straight ahead
<code>t.right(90)</code>	Turn right 90°	↓ Rotates clockwise
<code>t.backward(50)</code>	Move backward 50 steps	← Moves backward without turning
<code>t.left(45)</code>	Turn left 45°	↶ Rotates counter-clockwise

Drawing Geometric Shapes

We can use repetition loops (`for i in range(...)`) to magically draw perfect shapes!

Drawing a Square

Each side is 100 steps, and we turn right 90 degrees at each corner!

4 sides + 4 turns

```
import turtle
t = turtle.Turtle()

for i in range(4):
    t.forward(100)
    t.right(90)
```

Drawing a Triangle

A triangle has three sides. Each turn is 120 degrees to keep them equal!

3 sides + 3 turns

```
import turtle
t = turtle.Turtle()

for i in range(3):
    t.forward(100)
    t.right(120)
```

Drawing a Rectangle

A rectangle has two long sides and two short sides. Try changing the numbers to see how it reshapes!

```
import turtle
t = turtle.Turtle()

for i in range(2):
    t.forward(200)
    t.right(90)
    t.forward(100)
    t.right(90)
```

Pen Customization & Control

Make your artwork bright, colorful, and unique using pen properties and speed controls!

Pen Up & Pen Down

Lift the pen to skip spaces without drawing lines on your canvas:

`t.penup()` – Lifts the pen up.

`t.forward(80)` – Leaves a gap.

Colors & Thickness

Choose any color you like (red, green, yellow, etc.) and modify line width:

`t.pencolor("blue")` – Changes line to blue.

`t.pensize(5)` – Makes line thicker or thinner.

`t.pendown()` – Puts it down to draw again.

Changing Turtle Speed

Control how fast your turtle moves across the screen using values:

- `t.speed(1)` → Super slow turtle!
- `t.speed(5)` → Medium speed pace.
- `t.speed(10)` → Zoom zoom, very fast!

Drawing Fun Repeating Patterns

Loops can repeat your commands like magic to make amazing designs:

Star Pattern Loop:

```
for i in range(5):  
    t.forward(100)  
    t.right(144)
```

Spiral Design Loop:

```
for i in range(36):  
    t.forward(i * 2)  
    t.right(90)
```

Coding Creative Challenges



Challenge 1: Draw Your Name with Shapes!

Use what you learned to construct the letters of your name using squares, triangles, and rectangles! Feel free to choose different colors for each shape to make it uniquely yours.

★ Challenge 2: Create a Colorful Star

Try drawing a beautiful star with different colors and sizes. *Hint: A star needs 5 sides — turn right 144 degrees each time!*

```
t.pencolor("red")
t.pensize(3)
for i in range(5):
    t.forward(100)
    t.right(144)
```

🌀 Challenge 3: Make a Pattern with Loops

How many complex shapes can you make with just one loop? Try this code out to see an amazing pattern develop:

```
import turtle
t = turtle.Turtle()
for i in range(36):
    t.forward(100)
    t.right(170)
```

| Summary: You Are a Turtle Artist!

What we learned today:

- **Python Turtle introduction** & understanding basic movement commands.
- Drawing basic **geometric shapes** (Squares, Triangles, Rectangles) using loops.
- Controlling **pen color, line thickness, up/down states, and turtle speeds**.
- Creating complex **spirals and repeating patterns** through creative programming.

Thank you for joining Python Turtle class! Keep practicing, stay creative, and see you next time, coders! 🚀